

Square-Root Parity Splitting at a Quadratic Band-Merging Map:

Coupled Critical-Value Boundary Layers and a Corrected Small-Noise Law

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Abstract

At the algebraic band-merging parameter of $f_u(x) = 1 - ux^2$, deterministic dynamics has two cyclic components and the zero-noise Markov operator has a parity eigenvalue -1 . Positive Gaussian noise makes the operator strongly mixing and moves this mode to a simple real resonance $\lambda_-(\sigma) > -1$. Previous dense computations on $0.01 \leq \sigma \leq 0.03$ gave the striking fit $1 + \lambda_-(\sigma) \asymp \sigma^{2/3}$. We show that this is a crossover, not the small-noise exponent.

Two boundary layers determine the actual splitting. At the repelling component boundary $r = u_c - 1$, the rescaled left parity observable is exactly

$$H(\xi) = -\operatorname{erf}(\kappa\xi), \quad \kappa = \sqrt{\frac{\lambda^2 - 1}{2}}, \quad \lambda = 2u_cr.$$

It solves the Gaussian dilation equation $\mathbb{E}H(-\lambda\xi + Z) = -H(\xi)$. A second layer is created at the critical value 1. If ρ_c is the central-component invariant density at the critical point, then the signed right eigenmeasure satisfies

$$-\sqrt{\sigma} g_{\sigma,-}(1 - \sigma\xi) \longrightarrow R(\xi) = \rho_c \int_0^\infty \frac{\phi(u_c q^2 - \xi)}{\Phi(u_c q^2)} dq.$$

Pairing the two layers gives the nonanalytic spectral law

$$\boxed{1 + \lambda_-(\sigma) = C_* \sqrt{\sigma} + o(\sqrt{\sigma})}, \quad C_* = \int_0^\infty R(\xi) \operatorname{erfc}(a\xi) d\xi > 0,$$

where $a = 2u_c\kappa/\lambda$. The exact integral evaluates numerically to

$$\rho_c = 0.562641254486572\dots, \quad C_* = 0.105258535936908\dots$$

Consequently the parity lifetime is asymptotic to $(C_*\sqrt{\sigma})^{-1}$, and the 2/3-scaled gap diverges like $C_*\sigma^{-1/6}$.

Sparse folded computations with up to 204800 states reach $\sigma = 10^{-4}$. They reproduce the old-window exponent 0.667416, but the local exponent falls to 0.51613 and $(1 + \lambda_-)/\sqrt{\sigma}$ moves toward the predicted constant. These computations audit the asymptotic mechanism; the square-root law follows from the spectral-stability and boundary-layer argument.

Keywords: random perturbation; transfer operator; parity resonance; boundary layer; quadratic map; metastable spectrum; critical value; nonanalytic eigenvalue splitting.

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1 Introduction

Noise destroys exact cyclic decompositions. The subtle question is how quickly. At a quadratic band-merging parameter, the deterministic attractor has two components exchanged by one iterate. The associated Markov operator therefore has peripheral eigenvalues 1 and -1 . Every positive Gaussian noise makes the transition kernel strictly positive, leaving only the Perron root on the unit circle. The former parity mode becomes a long-lived negative resonance.

The preceding long-cycle study isolated this resonance and found

$$1 + \lambda_-(\sigma) \approx 0.2881505 \sigma^{0.6674161} \quad (1)$$

over its five smallest noise levels $0.01 \leq \sigma \leq 0.03$ [10]. It explicitly kept the $2/3$ exponent outside the theorem layer. Subsequent work completed the deterministic side: an analytic circle lift exposed the postcritical factors, and validated Wiener–Taylor bounds proved their noncancellation [11, 12]. Thus the stochastic splitting in (1) is now the remaining parity question.

The main point of this paper is that a single local layer gives the wrong answer. Smoothing the component jump at the repelling fixed point produces an exact error-function profile, but that profile is locally balanced and does not split the eigenvalue at leading order. Splitting occurs one preimage farther away, at the endpoint $x = 1$ that maps exactly onto the component boundary. The invariant density there is itself a noise-rounded square-root spike created by the critical point. Its height is $\sigma^{-1/2}$ and its width is σ ; their product forces the $\sqrt{\sigma}$ law.

Main results and status

- R1.** Standard stochastic spectral stability for the postcritically finite quadratic map gives a unique simple real resonance $\lambda_-(\sigma) \rightarrow -1$.
- R2.** The component-boundary observable has the exact universal profile $H(\xi) = -\operatorname{erf}(\kappa\xi)$. No fitted parameter enters κ .
- R3.** The critical point and the truncation normalizer produce a second, explicit endpoint-density profile R . The factor $\Phi(u_c q^2)^{-1}$ in R is essential; dropping row normalization changes the leading constant.
- R4.** A solvability pairing between the left profile H and the signed right density R proves

$$1 + \lambda_-(\sigma) = C_* \sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* > 0.$$

- R5.** Hence the parity e-folding time is $(C_* \sqrt{\sigma})^{-1}(1 + o(1))$. For $m\sqrt{\sigma} \rightarrow t$, the peripheral mode has the crossover $(-1)^m e^{-C_* t}$.
- R6.** The former $2/3$ law is ruled out asymptotically: $(1 + \lambda_-)/\sigma^{2/3} \rightarrow \infty$. Its accurate appearance on the old window is reproduced exactly and explained as preasymptotic drift.
- R7.** All finite matrices, fitted exponents, and profile comparisons are diagnostics. The decimal evaluation of C_* is numerical, while its positive integral formula and the exponent $1/2$ are analytic conclusions.

2 The folded Gaussian operator and deterministic parity

Let $u = u_c$ be the real root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (2)$$

and put

$$r = u_c - 1, \quad \lambda = 2u_cr. \quad (3)$$

Then

$$0 \mapsto 1 \mapsto -r \mapsto r \mapsto r, \quad (4)$$

and

$$u_c = 1.543689012692076\dots, \quad r = 0.543689012692076\dots, \quad \lambda = 1.678573510428322\dots \quad (5)$$

Write $f = f_{u_c}$ and fold the signed state by $x \mapsto |x|$. On $[0, 1]$ the deterministic map is

$$T(x) = |1 - u_c x^2|. \quad (6)$$

Up to the common endpoint r , the intervals

$$B = [0, r], \quad C = [r, 1] \quad (7)$$

are exchanged by T . Thus

$$h_0 = \mathbf{1}_B - \mathbf{1}_C, \quad h_0 \circ T = -h_0 \quad (8)$$

away from the two boundary preimages.

Let

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}, \quad Z_\sigma(x) = \int_{-1}^1 \phi_\sigma(y - f(x)) dy. \quad (9)$$

The normalized folded Markov operator acting on observables is

$$(\mathcal{K}_\sigma h)(x) = \frac{1}{Z_\sigma(x)} \int_{-1}^1 \phi_\sigma(y - f(x)) h(|y|) dy. \quad (10)$$

It is compact and strongly positive for every $\sigma > 0$. Its dual density operator is denoted by $\mathcal{P}_\sigma$.

2.1 Peripheral spectral stability

The map is postcritically finite and Misiurewicz. Its deterministic transfer operator is quasi-compact on the standard tower/spike space $\mathcal{B}$: a regular bounded-variation part is augmented by the finitely generated one-sided square-root singularities along (4) [1, 6, 7]. We use L^1 as the weak norm. The two mixing components give two simple peripheral eigenvalues 1 and -1 .

For completeness, we check the only feature of the present perturbation that is not covered verbatim by the usual additive-noise construction. Before folding, choose a slightly larger ambient interval $J \supset [-1, 1]$ with $f(J) \subset \text{int } J$; the ordinary tower estimates are applied there. Let Q_σ denote Gaussian convolution after extension by zero, and put $M_\sigma g = Z_\sigma^{-1} g$. Restriction and conditioning on the physical state interval then have the exact factorization

$$\mathcal{P}_\sigma g = \mathbf{1}_{[-1, 1]} Q_\sigma \mathcal{P}_0(M_\sigma g). \quad (11)$$

On a fixed neighborhood of the critical point,

$$Z_\sigma(x) = \Phi\left(\frac{u_c x^2}{\sigma}\right) + O(e^{-c/\sigma^2}). \quad (12)$$

Here Φ is the standard normal distribution function. Consequently $1 \leq M_\sigma \leq 2 + o(1)$, $\text{var } M_\sigma = O(1)$, and, because the regular part of $\mathcal{B}$ is bounded near $x = 0$,

$$\|(M_\sigma - \mathbf{I})g\|_{L^1} \leq C\sqrt{\sigma} \|g\|_{\mathcal{B}}, \quad \|M_\sigma g\|_{\mathcal{B}} \leq C\|g\|_{\mathcal{B}}. \quad (13)$$

Indeed, outside $|x| = O(\sqrt{\sigma})$ the first difference has a Gaussian tail in x^2/σ . The tower proof of the uniform Lasota–Yorke estimate uses only critical nonrecurrence and derivative growth; topological mixing is used later only to remove nontrivial peripheral roots [2]. Inserting (13) leaves that proof unchanged. The compact-support estimates extend to the Gaussian kernel by applying their variation bounds on dyadic tail shells and summing the exponentially decaying shell masses. Thus, for some $A, B < \infty$ and $\vartheta < 1$, uniformly for small σ ,

$$\|\mathcal{P}_\sigma^n g\|_{\mathcal{B}} \leq A\vartheta^n \|g\|_{\mathcal{B}} + B\|g\|_{L^1}, \quad \|\mathcal{P}_\sigma - \mathcal{P}_0\|_{\mathcal{B} \rightarrow L^1} = o(1). \quad (14)$$

These are precisely the strong/weak hypotheses of the Keller–Liverani theorem [5]. Notice that the order-one limit of M_σ at $x = 0$ is weakly small but cannot be discarded in the boundary layer calculation below.

Proposition 2.1 (Stable negative peripheral resonance). *There exist $\sigma_0 > 0$ and $\epsilon > 0$ such that, for $0 < \sigma < \sigma_0$, the spectrum of $\mathcal{K}_\sigma$ in $|z + 1| < \epsilon$ consists of one simple real eigenvalue $\lambda_-(\sigma)$. Moreover*

$$\lambda_-(\sigma) \longrightarrow -1. \quad (15)$$

There is a real signed eigenmeasure $g_\sigma(x) dx$ of $\mathcal{P}_\sigma$, normalized as below, which converges in L^1 to the deterministic signed component density and locally uniformly away from the postcritical set.

Proof. On the tower/spike space, -1 is isolated and simple because the square map is mixing on each component. The analytic circle realization and its validated reduced-sector gap provide, for this parameter, an explicit isolation of the component spectrum [12]. The uniform Lasota–Yorke inequality and weak convergence in (14), together with the compact strong-to-weak embedding, give stability of its rank-one spectral projector. A simple eigenvalue of a real operator remains real as long as it remains isolated: otherwise its complex conjugate would be a second eigenvalue in the same disk. Applying the convergent projector to g_0 and fixing the pairing normalization below gives $g_\sigma \rightarrow g_0$ in L^1 . On compact sets avoiding the finite postcritical set, all relevant inverse branches have derivative bounded away from zero. The local transfer equation then upgrades the regular part to uniform convergence; in particular, for some $\delta > 0$,

$$\sup_{0 \leq x \leq \delta} |g_\sigma(x) - \rho_D(x)| \longrightarrow 0. \quad (16)$$

□

Near a postcritical spike, the same tower bound gives, after reducing δ if necessary,

$$|g_\sigma(r + \sigma\xi)| \leq C\sigma^{-1/2}(1 + |\xi|)^{-1/2}, \quad |\xi| \leq \delta/\sigma. \quad (17)$$

This estimate and (16) are the two local consequences used below.

3 The deterministic central density

Let

$$S = f^2|_{[-r, r]}. \quad (18)$$

This is a mixing full-branch map on the central component. Its cosine lift $\pi(\theta) = -r \cos \theta$ is an analytic expanding circle map. Let $v(x)$ be the even analytic invariant density in interval coordinates, normalized by

$$\frac{1}{2\pi} \int_0^{2\pi} v(-r \cos \theta) d\theta = 1. \quad (19)$$

The exact Wiener–Taylor construction of the preceding two papers gives $\mathcal{T}_1 v = v$ and proves that this Perron eigenvalue is simple [11, 12].

Pushing normalized circle measure through the two-sheeted cover gives the central-component probability density

$$\rho_D(x) = \frac{v(x)}{\pi\sqrt{r^2 - x^2}}, \quad -r < x < r. \quad (20)$$

It is even, integrates to one, is analytic at the critical point $x = 0$, and has the expected square-root endpoint spikes. Define

$$\rho_c := \rho_D(0) = \frac{v(0)}{\pi r} > 0. \quad (21)$$

The Taylor implementation solves the normalized Perron equation at degrees 80, 120, 160 and obtains the identical displayed values

$$v(0) = 0.961019061704572\dots, \quad \rho_c = 0.562641254486572\dots \quad (22)$$

These decimals are used only to evaluate the final integral; the theorem uses the exact quantity in (21).

For later normalization, the zero-noise signed parity density on the folded space is

$$g_0(x) = \begin{cases} \rho_D(x), & x \in B, \\ -\frac{1}{2}\rho_C(x), & x \in C, \end{cases} \quad (23)$$

where $\rho_C = f_*\rho_D$ is the normalized density on the outer component. Evenness of ρ_D and normalization of ρ_C give

$$\int_0^r \rho_D(x) dx = \frac{1}{2}, \quad \int_r^1 \rho_C(x) dx = 1. \quad (24)$$

Thus both pieces in (23) have total absolute mass 1/2, and

$$\int_0^1 h_0(x)g_0(x) dx = 1. \quad (25)$$

We normalize g_σ by continuation of (25).

4 The repelling-boundary eigenprofile

Let Z be standard normal and define

$$\kappa = \sqrt{\frac{\lambda^2 - 1}{2}}, \quad H(\xi) = -\operatorname{erf}(\kappa\xi). \quad (26)$$

Then $H(-\infty) = 1$, $H(+\infty) = -1$, so it smooths the hard parity jump in the correct orientation.

Lemma 4.1 (Exact Gaussian dilation profile). *For every $\xi \in \mathbb{R}$,*

$$\mathbb{E} H(-\lambda\xi + Z) = -H(\xi). \quad (27)$$

Proof. The elementary Gaussian identity

$$\mathbb{E} \operatorname{erf}(A + BZ) = \operatorname{erf}\left(\frac{A}{\sqrt{1 + 2B^2}}\right) \quad (28)$$

follows by differentiating both sides in A and matching the value at zero. Since $1 + 2\kappa^2 = \lambda^2$, substituting $A = -\kappa\lambda\xi$ and $B = \kappa$ proves (27). $\square$

Use the global approximate parity observable

$$\hat{h}_\sigma(x) = H\left(\frac{x-r}{\sigma}\right). \quad (29)$$

Near r , write $x = r + \sigma\xi$. Since

$$f(r + \sigma\xi) = r - \lambda\sigma\xi - u_c\sigma^2\xi^2, \quad (30)$$

the full-Gaussian leading part of $\mathcal{K}_\sigma\hat{h}_\sigma$ is exactly the left side of (27). The row normalizer tends to one exponentially fast because r is an interior target. Hence the apparent $O(1)$ smoothing at r is locally balanced and contributes no leading eigenvalue defect.

The other preimage of the component boundary is the endpoint $x = 1$. Put $x = 1 - \sigma\xi$, $\xi \geq 0$. Then

$$f(1 - \sigma\xi) = -r + 2u_c\sigma\xi + O(\sigma^2\xi^2). \quad (31)$$

After folding, the target boundary coordinate is $-2u_c\xi - Z + o(1)$. Applying lemma 4.1 gives a second exact Gaussian identity.

Lemma 4.2 (Endpoint solvability defect). *Let*

$$a = \frac{2u_c\kappa}{\lambda}. \quad (32)$$

For every fixed $\xi \geq 0$,

$$(\mathcal{K}_\sigma\hat{h}_\sigma + \hat{h}_\sigma)(1 - \sigma\xi) \longrightarrow -\operatorname{erfc}(a\xi). \quad (33)$$

If $D_\sigma = \mathcal{K}_\sigma\hat{h}_\sigma + \hat{h}_\sigma$, there are $\delta, C, c > 0$ such that

$$|D_\sigma(r + \sigma\xi)| \leq C\sigma(1 + \xi^2)e^{-c\xi^2} + Ce^{-c/\sigma^2}, \quad |\xi| \leq \delta/\sigma, \quad (34)$$

$$|D_\sigma(1 - \sigma\xi)| \leq Ce^{-c\xi^2} + Ce^{-c/\sigma^2}, \quad 0 \leq \xi \leq \delta/\sigma. \quad (35)$$

Outside these two fixed δ -neighborhoods, $D_\sigma = O(e^{-c/\sigma^2})$.

Proof. By (31),

$$(\mathcal{K}_\sigma\hat{h}_\sigma)(1 - \sigma\xi) \longrightarrow \mathbb{E}H(-2u_c\xi - Z) = \operatorname{erf}\left(\frac{2u_c\kappa}{\lambda}\xi\right).$$

Meanwhile $\hat{h}_\sigma(1 - \sigma\xi) \rightarrow -1$. Their sum is $-\operatorname{erfc}(a\xi)$. The remaining estimates follow from Gaussian tails and the Taylor remainders in (30) and (31). More explicitly, convolution with H' turns the quadratic displacement $u_c\sigma\xi^2$ in the rescaled r -coordinate into the first bound in (34); the convolved derivative has Gaussian decay in ξ . At the endpoint and on the complement, the distance of the image from the parity boundary gives (35) and the stated off-layer estimate. $\square$

Numerically,

$$\kappa = 0.953312391063827\dots, \quad a = 1.753414854465239\dots \quad (36)$$

5 The critical-value endpoint density

The defect in lemma 4.2 must be paired with the signed right eigenmeasure near $x = 1$. That density cannot be replaced by its formal zero-noise spike at the scale $1 - x = O(\sigma)$, because the critical source and the truncation normalizer are resolved on exactly the same scale.

Let $\phi(t) = (2\pi)^{-1/2}e^{-t^2/2}$ and let Φ be its distribution function. Set

$$R(\xi) = \rho_c \int_0^\infty \frac{\phi(u_c q^2 - \xi)}{\Phi(u_c q^2)} dq, \quad \xi \geq 0. \quad (37)$$

Lemma 5.1 (Critical endpoint layer). *For each compact interval of $\xi \geq 0$,*

$$-\sqrt{\sigma} g_\sigma(1 - \sigma\xi) \longrightarrow R(\xi) \quad (38)$$

locally uniformly. Moreover there is an integrable majorant after multiplication by the endpoint defect: for some $\delta, C > 0$,

$$\sqrt{\sigma} |g_\sigma(1 - \sigma\xi)| \leq C(1 + \xi)^{-1/2}, \quad 0 \leq \xi \leq \delta/\sigma. \quad (39)$$

In addition,

$$R(\xi) \sim \frac{\rho_c}{2\sqrt{u_c\xi}} \quad (\xi \rightarrow \infty). \quad (40)$$

Proof. The only deterministic sources that can reach $1 - \sigma\xi$ lie at $x = O(\sqrt{\sigma})$ in the central component. Write $x = \sqrt{\sigma}q$. Then

$$f(x) = 1 - u_c\sigma q^2, \quad Z_\sigma(x) = \Phi(u_cq^2) + O(e^{-c/\sigma^2}). \quad (41)$$

At $y = 1 - \sigma\xi$, the transition density is therefore

$$\frac{1}{\sigma} \frac{\phi(u_cq^2 - \xi)}{\Phi(u_cq^2)} (1 + o(1)). \quad (42)$$

More precisely, the folded eigenrelation gives

$$-\sqrt{\sigma} g_\sigma(y) = \frac{\sqrt{\sigma}}{-\lambda_-(\sigma)} \int_0^1 \frac{g_\sigma(x)}{Z_\sigma(x)} \{ \phi_\sigma(y - f(x)) + \phi_\sigma(-y - f(x)) \} dx. \quad (43)$$

By (16), $g_\sigma(\sqrt{\sigma}q) \rightarrow \rho_D(0) = \rho_c$ uniformly on compact q -sets. The second Gaussian in (43) and all sources outside a fixed central neighborhood are exponentially small. Using $dx = \sqrt{\sigma} dq$ in the first Gaussian, and $-\lambda_-(\sigma) \rightarrow 1$, proves (38).

For the uniform bound, choose δ so that every source of $[1 - \delta, 1]$ belongs to the central neighborhood on which g_σ is uniformly bounded, apart from exponentially unlikely Gaussian jumps. Split the q -integral at $q = 2\sqrt{(1 + \xi)/u_c}$ and use Gaussian decay on the outer part; on the inner part $\Phi(u_cq^2) \geq 1/2$. The same change of variables as above gives (39). Applied to R , it gives $R(\xi) \leq C(1 + \xi)^{-1/2}$. Finally, Laplace expansion about $q = \sqrt{\xi/u_c}$ yields (40). Together with (35), the product needed below has an integrable majorant. $\square$

The denominator in (37) has a direct probabilistic meaning. At the critical scale, an untruncated Gaussian centered at $1 - u_c\sigma q^2$ remains inside the upper state boundary with probability $\Phi(u_cq^2)$. The Markov kernel conditions on that event. Replacing the denominator by one changes the leading constant and is therefore not a harmless normalization simplification.

6 Square-root splitting and parity lifetime

The two layers now meet through a one-line spectral solvability identity. For the signed eigenmeasure $g_\sigma dx$,

$$\begin{aligned} (\lambda_-(\sigma) + 1) \int_0^1 \hat{h}_\sigma(x) g_\sigma(x) dx \\ = \int_0^1 (\mathcal{K}_\sigma \hat{h}_\sigma + \hat{h}_\sigma)(x) g_\sigma(x) dx. \end{aligned} \quad (44)$$

The denominator on the left tends to one by (25). The endpoint $x = 1$ supplies the only $\sqrt{\sigma}$ contribution on the right.

Theorem 6.1 (Square-root parity splitting). *As $\sigma \downarrow 0$,*

$$\boxed{1 + \lambda_-(\sigma) = C_*\sqrt{\sigma} + o(\sqrt{\sigma})}, \quad (45)$$

where

$$C_* = \int_0^\infty R(\xi) \operatorname{erfc}(a\xi) d\xi \quad (46)$$

$$= \rho_c \int_0^\infty \frac{dq}{\Phi(u_c q^2)} \int_0^\infty \phi(u_c q^2 - \xi) \operatorname{erfc}(a\xi) d\xi. \quad (47)$$

In particular $0 < C_* < \infty$.

Proof. In (44), split the state interval into fixed δ -neighborhoods of r and 1 and their complement. On the complement, the residual in lemma 4.2 is exponentially small, while $\|g_\sigma\|_{L^1}$ is uniformly bounded. In the r -layer, put $x = r + \sigma\xi$. From (34) and (17),

$$\begin{aligned} \int_{|x-r|\leq\delta} |D_\sigma(x)g_\sigma(x)| dx &\leq C\sigma^{3/2} \int_{\mathbb{R}} (1 + \xi^2)(1 + |\xi|)^{-1/2} e^{-c\xi^2} d\xi + O(e^{-c/\sigma^2}) \\ &= O(\sigma^{3/2}). \end{aligned}$$

This also controls the rescaled tails that are invisible in a merely local boundary-layer expansion.

In the endpoint layer put $x = 1 - \sigma\xi$. By lemmas 4.2 and 5.1,

$$\begin{aligned} (\mathcal{K}_\sigma \hat{h}_\sigma + \hat{h}_\sigma)(x)g_\sigma(x) dx \\ = \sqrt{\sigma} R(\xi) \operatorname{erfc}(a\xi) d\xi + o(\sqrt{\sigma}). \end{aligned}$$

The majorant in lemma 5.1 permits dominated convergence. This yields $C_*\sqrt{\sigma} + o(\sqrt{\sigma})$ on the right of (44). Its left pairing tends to one. Positivity and finiteness follow immediately from (47). $\square$

Numerical Taylor solution of the exact component-density equation and nested quadrature give

$$\boxed{C_* = 0.105258535936908 \dots} \quad (48)$$

The integral formula, not the displayed decimal, is the theorem.

Corollary 6.2 (Failure of the two-thirds asymptotic). *The old scaling satisfies*

$$\frac{1 + \lambda_-(\sigma)}{\sigma^{2/3}} = C_*\sigma^{-1/6}(1 + o(1)) \longrightarrow +\infty. \quad (49)$$

Thus no finite nonzero 2/3-law constant exists.

Corollary 6.3 (Parity lifetime and crossover). *Let*

$$m_{\text{par}}(\sigma) = -\frac{1}{\log |\lambda_-(\sigma)|}. \quad (50)$$

Then

$$m_{\text{par}}(\sigma) \sim \frac{1}{C_*\sqrt{\sigma}}. \quad (51)$$

If $m = m(\sigma)$ has fixed parity and $m\sqrt{\sigma} \rightarrow t \in [0, \infty)$, then

$$\lambda_-(\sigma)^m \longrightarrow (-1)^m e^{-C_* t}. \quad (52)$$

Proof. Write $\lambda_-(\sigma) = -(1 - C_*\sqrt{\sigma} + o(\sqrt{\sigma}))$ and expand the logarithm. Both conclusions follow. $\square$

The crossover concerns the isolated parity factor. A statement replacing the full trace by $1 + \lambda_-^m$ additionally requires uniform control of all bulk resonances as $\sigma \rightarrow 0$; that separate assertion is not used here.

7 Sparse spectral and profile audit

The numerical work has three purposes: extend the old noise window, test the two local profiles independently, and verify spatial resolution. It is not used to infer the exponent in [theorem 6.1](#).

For midpoint nodes $x_i = (i + 1/2)/d$ on $[0, 1]$, the folded row weights are

$$K_{ij} \propto \exp\left[-\frac{(x_j - f(x_i))^2}{2\sigma^2}\right] + \exp\left[-\frac{(-x_j - f(x_i))^2}{2\sigma^2}\right]. \quad (53)$$

Only destinations within eight standard deviations of $|f(x_i)|$ are stored; the rows are then normalized exactly. Keeping $d\sigma \simeq 20.48$ makes the number of nonzeros linear in d . ARPACK resolves the largest-modulus eigenvalues. NumPy, SciPy, and Matplotlib are used for the implementation [3, 4, 8].

7.1 The exponent crossover

[Table 1](#) gives representative values. The row-sum error is at most 5.6×10^{-16} throughout.

Table 1: Negative parity resonance across the extended noise window.

σ	d	$1 + \lambda_-$	$(1 + \lambda_-)/\sqrt{\sigma}$	$(1 + \lambda_-)/\sigma^{2/3}$	local power
3×10^{-2}	683	2.80506×10^{-2}	0.161950	0.290532	—
1×10^{-2}	2048	1.34466×10^{-2}	0.134466	0.289698	0.62787
5×10^{-3}	4096	8.84596×10^{-3}	0.125101	0.302528	0.57272
2×10^{-3}	10240	5.26120×10^{-3}	0.117644	0.331435	0.56022
1×10^{-3}	20480	3.61072×10^{-3}	0.114181	0.361072	0.54465
5×10^{-4}	40960	2.50024×10^{-3}	0.111814	0.396889	0.53022
2×10^{-4}	102400	1.54905×10^{-3}	0.109534	0.452945	0.52248
1×10^{-4}	204800	1.08316×10^{-3}	0.108316	0.502746	0.51613

A log-log fit on exactly the old five-point window $0.01 \leq \sigma \leq 0.03$ reproduces

$$p_{\text{old}} = 0.6674161083, \quad A_{\text{old}} = 0.2881504866. \quad (54)$$

On $10^{-4} \leq \sigma \leq 0.002$, the fitted power is already

$$p_{\text{new}} = 0.52749 \dots, \quad (55)$$

and the final two-point local exponent is 0.51613. More decisively, the 2/3-scaled column grows monotonically while the square-root-scaled column moves toward (48).

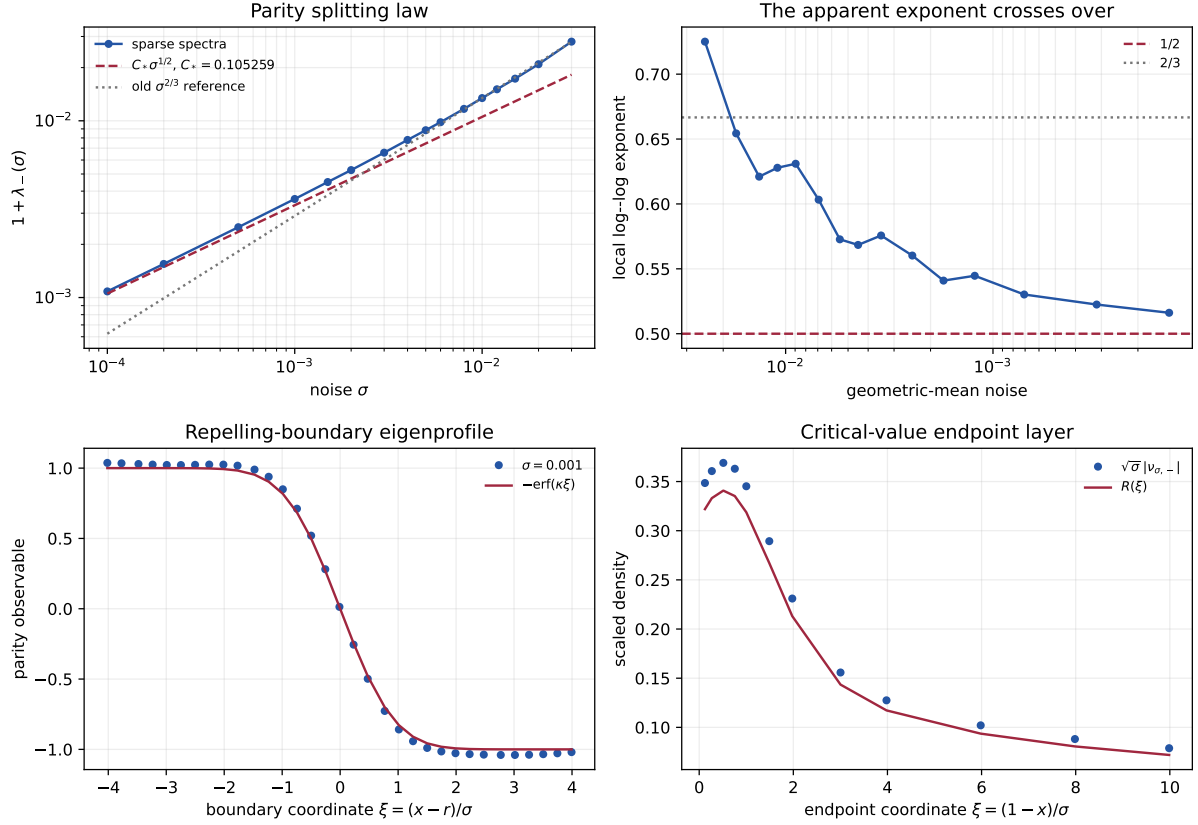


Figure 1: Square-root boundary-layer audit. Top left: the extended parity gap and the analytic square-root law; the dotted $2/3$ reference only follows the intermediate window. Top right: local exponents drift from the old fit toward $1/2$. Bottom left: the right matrix eigenvector collapses onto the exact error-function boundary profile. Bottom right: the left signed eigenvector density approaches the independently derived critical-value profile R .

7.2 Resolution and profile checks

At $\sigma = 0.001$, increasing $d\sigma$ from 10.24 to 30.72 changes the negative eigenvalue by only 6.40×10^{-6} , or 0.18% of the parity gap. The value at $d\sigma = 20.48$ differs from the finest one by 9.98×10^{-7} .

The lower panels of [figure 1](#) test the two ingredients of the proof separately at $\sigma = 0.001$, $d = 20480$. The observed parity observable differs from H by at most about 3.7% on $|\xi| \leq 4$. The endpoint signed density has an 8–10% preasymptotic offset but follows the full shape of R , including its nonmonotone maximum. Neither profile is fitted to the matrix data: κ , a , ρ_c , and the row-normalization factor are fixed before diagonalization.

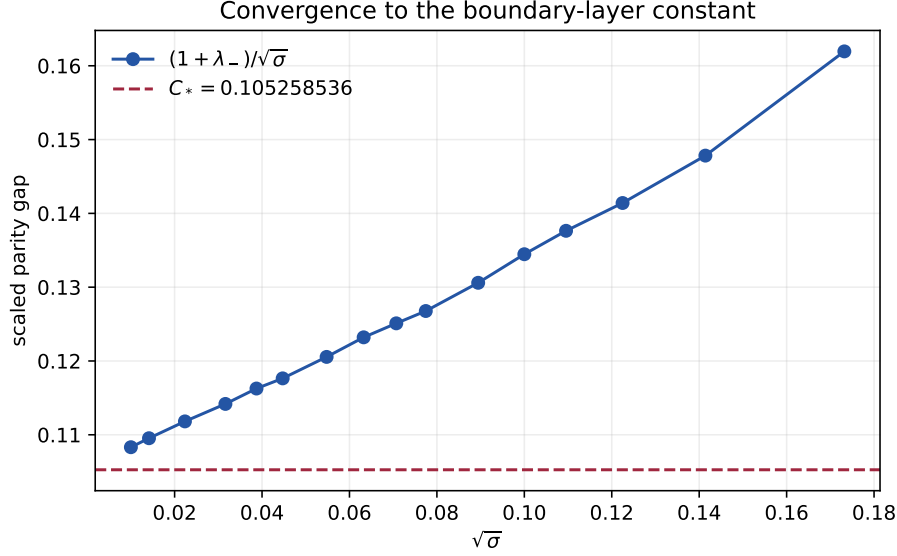


Figure 2: The scaled gap $(1 + \lambda_-) / \sqrt{\sigma}$ approaches the boundary-layer constant C_* . The horizontal value comes from (47), not from regression against the resonance data.

7.3 Reproducibility and numerical status

The sparse calculation at $\sigma = 10^{-4}$ has 204800 states and 6.77×10^7 nonzeros. The archived output contains all sixteen noise levels, five independent resolutions, both profile tables, software versions, and source hashes. The test suite checks the sparse/dense spectral match, the Taylor stability of ρ_c , the exact Gaussian eigenprofile identity, the nested integral for C_* , and a finite-noise parity resonance.

The quadrature and matrix eigenvalues are ordinary floating-point computations. They support the displayed decimals and figures but are not a computer-assisted proof. The theorem asserts an exact positive integral constant and the exponent $1/2$; neither assertion depends on a rounded eigenvalue or fitted slope.

8 Discussion

8.1 Why the exponent is one half

The critical source scale is $x = O(\sqrt{\sigma})$ because $1 - f(x) = u_c x^2$. It creates an endpoint density of height $\sigma^{-1/2}$ over a layer of width σ . The endpoint then maps to the parity boundary, where the error-function profile has an $O(1)$ solvability defect. Thus

$$\underbrace{\sigma^{-1/2}}_{\text{critical density}} \times \underbrace{\sigma}_{\text{endpoint width}} \times \underbrace{1}_{\text{parity defect}} = \sqrt{\sigma}. \quad (56)$$

The exponent is therefore geometric, not a free regression parameter.

The repelling boundary r is essential but does not itself generate the splitting. Its local Gaussian dilation problem is exactly solvable at eigenvalue -1 . The gap appears because the second preimage $x = 1$ ends at the state boundary and is fed by a quadratic critical point. This is why a one-layer leakage estimate misses both the constant and, over moderate noise, the apparent exponent.

8.2 The corrected two-clock picture

The deterministic periodic-orbit localization method has the logarithmic ceiling

$$m_{\text{loc}}(\sigma) \lesssim \frac{\log(1/\sigma)}{\log \lambda} \quad (57)$$

because some cycles approach the state boundary exponentially [10]. The parity lifetime is now

$$m_{\text{par}}(\sigma) \sim C_*^{-1} \sigma^{-1/2}. \quad (58)$$

It remains parametrically longer than (57), although shorter than the previously conjectured $\sigma^{-2/3}$ scale. At $\sigma = 10^{-4}$, the resolved e-folding time is about 923, while the leading formula gives 950 and the logarithmic orbit ceiling is only about 17.8.

This separation is constructive. It says that a simultaneous small-noise/long-cycle determinant cannot be obtained by extending isolated periodic Gaussian tubes to the parity lifetime. The correct intermediate description must first extract the factor $1 - z\lambda_-(\sigma)$ and treat the two coupled boundary layers before attempting a bulk determinant limit.

9 Conclusion

The noisy parity resonance at the quadratic band-merging parameter has a nonanalytic but explicit small-noise splitting. The component-boundary observable is an exact error function, the critical endpoint density is a normalized Gaussian–quadratic integral, and their solvability pairing gives

$$1 + \lambda_-(\sigma) = 0.105258535936908 \dots \sqrt{\sigma} + o(\sqrt{\sigma}).$$

The formerly observed 2/3 exponent is a genuine and reproducible intermediate-window fit, but it is not the asymptotic law.

This closes the stochastic parity question left by the long-cycle analysis: the negative resonance converges to -1 , its lifetime scale is $\sigma^{-1/2}$, and the crossover profile is explicit. The next spectral gate is no longer an exponent fit. It is to determine whether the parity-extracted noisy bulk determinant has a target-independent scaling or scattering limit.

Data and code availability

The source, tests, sparse spectra, resolution audit, boundary profiles, figures, and machine-readable summary are archived with this paper [9].

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